

THE MEDIUM — AN ATTEMPTED DESCRIPTION

**Deriving the Gravitational Constant  $G$   
and the Hubble Constant  $H_0$**

from Negotiation Substrate Primitives

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Zenodo DOI: 10.5281/zenodo.18927154

[github.com/bkdepuy/The-Medium](https://github.com/bkdepuy/The-Medium)

2026

# 1 Abstract

This paper derives the gravitational constant  $G$  and the Hubble constant  $H_0$  within The Medium, using only the foundational definition adopted in the companion papers:  $\iota$  size 1 = Planck length.  $G$  is derived from the Friedmann equation using the vacuum floor density  $\rho_\Lambda$  and the Hubble rate, achieving 94–97%

## *Key result*

### Scale inputs (from Document 1):

Input 1:  $\iota$  size 1 = Planck length =  $1.616 \times 10^{-35}$  m (the substrate grain)

Input 2:  $H_0 = 68.6$  km/s/Mpc (the current Hubble constant, identified as the second scale input)

In Planck units:  $G = 1$ ,  $c = 1$ ,  $\hbar = 1/(2\pi)$ .

# 2 Part I: Deriving the Gravitational Constant $G$

## *In plain language*

$G$  is the number that tells you how strongly two masses attract each other. In standard physics it is simply measured — nobody knows why it has the value it does.

The Medium’s approach: dimensional analysis applied to the two substrate scale quantities ( $\rho_\Lambda$  and  $H_0$ ) produces the unique combination with the units of  $G$ . The 94–97% numerical agreement with the measured value is non-trivial given that these quantities span many orders of magnitude. The gap of 3–6% has not been derived; a correction of the right magnitude exists in loop quantum gravity (the Barbero-Immirzi parameter), but connecting it to the substrate dynamics is an open calculation, not an established result.

## 2.1 What $G$ Is in The Medium

In general relativity,  $G$  appears in the Einstein field equations as the coupling between spacetime curvature and energy-momentum. In The Medium, curvature is the local variation in negotiation graph density.  $G$  is therefore the medium’s compliance to gravitational perturbations — how readily the strain field responds to a mass imperfection:

$$\nabla^2 \Phi = 4\pi G \rho \quad (\text{Poisson’s equation for the gravitational strain field}) \quad (1)$$

## 2.2 Dimensional Argument

$G$  has SI units  $\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$ . The available quantities:

$$\begin{aligned} [\rho_\Lambda] &= \text{kg m}^{-3} \\ [H_0] &= \text{s}^{-1} \\ \left[ \frac{H_0^2}{\rho_\Lambda} \right] &= \frac{\text{s}^{-2}}{\text{kg m}^{-3}} = \text{m}^3 \text{kg}^{-1} \text{s}^{-2} \end{aligned}$$

$H_0^2/\rho_\Lambda$  is the unique combination of the vacuum floor and Hubble rate that carries the units of  $G$ . This is a result of dimensional analysis, not a derivation from substrate dynamics. The factor  $4\pi$  appears in Poisson's equation from Gauss's theorem in three dimensions; it is fixed by the geometry of the projection, not a free parameter. The physical interpretation —  $\rho_\Lambda$  as medium stiffness,  $H_0^2$  as dynamic factor,  $G$  as compliance — is a consistent reading but must be validated by deriving the strain field equations from the substrate. That derivation does not yet exist.

### 2.3 The Result

$$\boxed{G = \frac{H_0^2}{4\pi\rho_\Lambda}} \quad (2)$$

Note on the relationship to the Friedmann equation: the full Friedmann equation at the present epoch is  $H_0^2 = (8\pi G/3)(\rho_m + \rho_\Lambda + \rho_r)$ . For a  $\Lambda$ -dominated flat universe,  $\rho_{\text{total}} \approx \rho_\Lambda$ , giving  $G \approx 3H_0^2/(8\pi\rho_\Lambda)$ . The Medium's formula  $G = H_0^2/(4\pi\rho_\Lambda)$  differs by a factor of  $3/2$ . This factor is the origin of the  $\sim 3\%$

### 2.4 Numerical Verification

Using Planck 2018 cosmological parameters ( $H_0 = 67.36 \text{ km/s/Mpc}$ ,  $\Omega_\Lambda = 0.685$ ):

$$\begin{aligned} H_0 &= 2.183 \times 10^{-18} \text{ s}^{-1} \\ \rho_\Lambda &= \Omega_\Lambda \times \rho_c = 5.836 \times 10^{-27} \text{ kg m}^{-3} \\ G_{\text{derived}} &= \frac{(2.183 \times 10^{-18})^2}{4\pi (5.836 \times 10^{-27})} = 6.498 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \\ G_{\text{measured}} &= 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \\ \text{Agreement} &= 97.4\% \end{aligned}$$

With WMAP9-era parameters ( $H_0 = 68.0$ ,  $\Omega_\Lambda = 0.71$ ): agreement = 94.0%

### 2.5 The Analytic Prediction

Substituting  $\rho_\Lambda = \Omega_\Lambda \rho_c$  and the Friedmann equation  $\rho_c = 3H_0^2/(8\pi G)$ :

$$G_{\text{derived}} = \frac{H_0^2}{4\pi \Omega_\Lambda \rho_c} = \frac{H_0^2}{4\pi \Omega_\Lambda} \cdot \frac{8\pi G}{3H_0^2} = \frac{2G}{3\Omega_\Lambda} \quad (3)$$

Therefore:

$$\frac{G_{\text{derived}}}{G} = \frac{2}{3\Omega_{\Lambda}} \quad \Longleftrightarrow \quad \boxed{\Omega_{\Lambda} = \frac{2}{3} \approx 0.667} \quad (4)$$

This is an exact algebraic result. Observed:  $\Omega_{\Lambda} \approx 0.685$  (Planck 2018). Discrepancy: 2.7\

#### *Testable prediction*

**Consequence of G formula:**  $\Omega_{\Lambda} = 2/3$

This follows algebraically from  $G = H_0^2/4\pi\rho_{\Lambda}$  substituted into the Friedmann equation. It is not an independent prediction — if the G formula is wrong, so is  $\Omega_{\Lambda} = 2/3$ . They stand or fall together. Current best value: 0.685, discrepancy 2.7%.

## 2.6 The Barbero-Immirzi Correction

The residual 3–6\

$$G_{\text{precise}} = G_{\text{derived}} \times f(\gamma), \quad f(0.2375) \approx 1.027\text{--}1.064 \quad (5)$$

The same fractional gap ( $\sim 6\%$ )

## 3 Part II: Deriving the Hubble Constant $H_0$

### *In plain language*

$H_0$  is the current expansion rate of the universe. It is one of the most contested numbers in modern cosmology: telescopes measuring nearby galaxies get  $\sim 73$  km/s/Mpc while the CMB gives  $\sim 67$  km/s/Mpc. Nobody agrees which is right.

$H_0$  is the second scale input — it is not derived from the substrate; it is the observed Hubble constant used as a scale anchor alongside the Planck length. The preferred value 68.6 km/s/Mpc is selected for internal consistency (it gives the best agreement with  $G$  via the Friedmann equation and is consistent with the proton mass via the geometric mean formula). The Hubble tension interpretation — that CMB and local measurements probe different aspects of the substrate — is a hypothesis, not a conclusion.

### 3.1 $H_0$ as the Second Scale Input

This description has exactly two inputs:

1. Iota scale:  $l_{\epsilon} = \text{Planck length (the substrate grain)}$
2. Hubble scale:  $H_0$  (the current cosmic horizon frequency)

This description has two inputs: the iota scale (Planck length) and  $H_0$ . Given these two inputs and the geometric mean formula,  $\epsilon$  follows. Given  $\epsilon$ , the proton mass follows. Given  $H_0$  and  $\rho_{\Lambda}$ ,  $G$  follows from the Friedmann equation. The preferred value  $H_0 = 68.6$  km/s/Mpc is the value that makes the three quantities  $G$ ,  $\epsilon$ , and  $m_{\text{proton}}$  mutually consistent at the observed values. It is selected for internal consistency, not derived from first principles.

### 3.2 The Hubble Floor Energy

$H_0$  is identified as the negotiation frequency at the Hubble floor: the oscillation rate of the longest wavelength negotiation mode the universe can sustain. The longest sustainable negotiation wavelength is the Hubble radius  $R_H = c/H_0$ . The Hubble floor energy:

$$E_{\text{Hubble}} = hH_0 = 6.626 \times 10^{-34} \times 2.222 \times 10^{-18} \text{ J} = 9.19 \times 10^{-52} \text{ J} = 9.19 \times 10^{-39} \text{ MeV} \quad (6)$$

### 3.3 The Hubble Floor Condition

The Hubble floor energy must equal the vacuum energy density times the Hubble volume:

$$E_{\text{Hubble}} = \rho_{\Lambda} \cdot V_{\text{Hubble}} = \rho_{\Lambda} \cdot \frac{4\pi}{3} R_H^3 \quad (7)$$

Substituting  $R_H = c/H_0$  and solving:

$$\boxed{H_0^4 = \frac{\rho_{\Lambda} \cdot 4\pi c^3}{3h}} \implies H_0 = \left[ \frac{4\pi c^3 \rho_{\Lambda}}{3h} \right]^{1/4} \quad (8)$$

### 3.4 Numerical Derivation of $H_0$

Working from the geometric mean formula (Document 2) with  $\varepsilon = 312.8 \text{ MeV}$ :

$$E_{\text{Hubble}} = \frac{\varepsilon^3}{E_{\text{Planck}}^2} \implies R_H = \frac{hc}{E_{\text{Hubble}}} \quad (9)$$

With the Barbero-Immirzi and gluon edge corrections applied to  $\varepsilon$ :

$$H_0 = c/R_H = 68.6 \text{ km/s/Mpc} \quad (10)$$

### 3.5 The Hubble Tension in The Medium

| Method                | Value         | Medium interpretation                                           |
|-----------------------|---------------|-----------------------------------------------------------------|
| CMB (Planck 2018)     | 67.4 km/s/Mpc | Global substrate negotiation frequency; most direct measurement |
| Local distance ladder | 73.0 km/s/Mpc | Local measurement biased upward by local overdensity            |
| Medium prediction     | 68.6 km/s/Mpc | Substrate negotiation frequency; 1.2 km/s/Mpc above CMB         |

#### *Testable prediction*

#### **Interpretation: Hubble tension**

$H_0 = 68.6 \text{ km/s/Mpc}$  is selected for internal consistency, not derived from the substrate. The interpretation that local measurements are biased upward by  $\sim 4.4 \text{ km/s/Mpc}$  due to local overdensity is a hypothesis without a derivation behind the bias claim. Whether large-volume surveys converge toward 68–69 km/s/Mpc is a consequence of that hypothesis, not an independent prediction.

## 4 Part III: The Cosmological Constant

### *In plain language*

The cosmological constant  $\Lambda$  is the energy density of empty space. Quantum field theory predicts it should be  $10^{123}$  times larger than what is observed. This is the worst unsolved problem in physics.

The Medium reframes it. The  $10^{123}$  hierarchy between the Planck-scale QFT prediction and the observed  $\Lambda$  is identified as the ratio of the two scale inputs: the depth of the Medium between its grain and its horizon. This reframing is only a resolution if the Hamiltonian floor at  $\rho_\Lambda$  is a genuine physical threshold — meaning the Hamiltonian is undefined below that density. That condition must be derived from substrate dynamics to close the argument. The reframing identifies where the problem lives; it does not yet dissolve it.

### 4.1 The Cosmological Constant in Planck Units

$$\Lambda_{\text{Planck}} = \Lambda_{\text{observed}} \cdot l_{\text{Planck}}^2 \approx 2.9 \times 10^{-122} \quad (11)$$

The Medium explains this as:

$$\Lambda_{\text{Planck}} \approx \left( \frac{l_{\text{Planck}}}{R_H} \right)^2 = \left( \frac{1.616 \times 10^{-35}}{1.350 \times 10^{26}} \right)^2 \approx 1.4 \times 10^{-122} \quad (12)$$

### 4.2 Why the Hierarchy Is Not a Problem

The  $10^{123}$  hierarchy between the QFT prediction and observed  $\Lambda$  arises because QFT uses the Planck scale as its vacuum reference. The Medium identifies the correct vacuum reference as the Hubble floor. In Planck units:

$$\frac{\rho_{\text{Planck}}}{\rho_\Lambda} = \frac{E_{\text{Planck}}/l_{\text{Planck}}^3}{E_{\text{Hubble}}/R_H^3} = \frac{E_{\text{Planck}}}{E_{\text{Hubble}}} \cdot \left( \frac{R_H}{l_{\text{Planck}}} \right)^3 \approx 10^{123} \quad (13)$$

This is the depth of the Medium — the dynamic range of the negotiation substrate from Planck grain to cosmic horizon. It is not a cancellation problem. It is a consequence of the universe being  $\sim 10^{60}$  Planck lengths across.

## 5 Part IV: Connections Between $G$ , $H_0$ , and the Proton Mass

### *In plain language*

The three derivations —  $G$  at 94%,  $H_0$  at 68.6 km/s/Mpc, and the proton mass within 10% — are not independent. They all flow from the same two inputs and the same correction factor. When three apparently unrelated quantities share a common derivation and a common residual correction, this description is pointing at genuine structure.

## 5.1 The Common Correction Factor

All three derivations have a residual gap closed by the same Barbero-Immirzi correction:

| Quantity                  | Gap                                                           | Correction                                 |
|---------------------------|---------------------------------------------------------------|--------------------------------------------|
| $G$                       | 3–6%                                                          | $f(\gamma) \approx 1.027\text{--}1.064$    |
| Proton mass $\varepsilon$ | 10.3%                                                         | $(2\pi)^{1/3}$ correction partially closes |
| $\hbar$                   | 0.9% (non-circular derivation; see $\hbar$ companion section) | $f(\gamma) \approx 1.009$                  |

In loop quantum gravity,  $\gamma \approx 0.2375$  is fixed by requiring that the Bekenstein-Hawking entropy formula is reproduced by counting quantum states of the area operator. The Medium independently requires  $\gamma$  to close three separate gaps. This convergence strongly suggests  $\gamma$  is not a free parameter in The Medium but is fixed by the same entropy counting.

## 5.2 The Consistency Triangle

The three quantities form a consistency triangle:

$$G \longleftrightarrow H_0 \longleftrightarrow m_{\text{proton}} \quad (14)$$

Given any two, the third is constrained by the Friedmann equation and the geometric mean formula:

- Given  $G$  and  $H_0$ :  $\rho_c \rightarrow E_{\text{Hubble}} \rightarrow \varepsilon \rightarrow m_{\text{proton}}$
- Given  $G$  and  $m_{\text{proton}}$ :  $\varepsilon \rightarrow E_{\text{Hubble}} \rightarrow H_0$
- Given  $H_0$  and  $m_{\text{proton}}$ :  $\varepsilon \rightarrow E_{\text{Hubble}} \rightarrow G$  via Friedmann

All three paths give mutually consistent results within the identified correction factors. This triangular consistency is strong evidence that this description is not fitting three separate numbers but is describing a single underlying structure with three observable manifestations.

## 5.3 Predictions

### *Testable prediction*

#### **Predictions from cosmological evolution:**

1.  $H_0 = 68.6 \pm 0.5$  km/s/Mpc — selected for internal consistency; convergence of large-volume surveys toward this value would be supporting evidence
2.  $\Lambda$  variation —  $d\Lambda/dt = -2\Lambda \cdot H_0$ ; no current observational constraint at this precision

### *Open calculation*

#### **Observational tensions — may rule out current evolution equations**

The cosmological evolution of  $G$  and  $m_{\text{proton}}$  implied by the two-input description conflicts with current observations:

- $\dot{G}/G \approx -1.5 \times 10^{-10}/\text{yr}$  (from  $dG/dt = -2G \cdot H_0$ ). Current bounds:  $|\dot{G}/G| < 4\text{--}9 \times 10^{-13}/\text{yr}$  (MESSENGER, LLR, pulsar timing) — roughly 200× tighter. This

rate appears observationally ruled out.

- $\dot{m}_p/m_p \approx 7 \times 10^{-11}/\text{yr}$  (from  $dm_p/dt = m_p \cdot H_0/3$ ). Current bounds:  $|\dot{m}_p/m_p| < 10^{-14}/\text{yr}$  (atomic clocks, quasar spectroscopy) — roughly 7000× tighter. This rate also appears observationally ruled out.

Resolving these tensions requires either a physical mechanism that suppresses local variation while allowing secular variation over Hubble timescales, or a revision of the cosmological evolution equations. Until one of these is identified, these are failed predictions of the current description.

## 6 Summary

### Key result

#### Results from two inputs:

| Quantity                     | Derived                             | Agreement                |
|------------------------------|-------------------------------------|--------------------------|
| $m_{\text{proton}}$          | $3\varepsilon = 938.3 \text{ MeV}$  | Exact (by construction)  |
| $G$                          | $H_0^2/4\pi\rho_\Lambda$            | 94–97%                   |
| $\hbar$                      | $m_p c k(\gamma) r_p/6\pi$          | 99.1%                    |
| $H_0$                        | 68.6 km/s/Mpc                       | Within Hubble tension    |
| $\Lambda$                    | $\approx 10^{-122}$ in Planck units | Same order               |
| $\Omega_\Lambda$ (predicted) | $2/3 = 0.667$                       | 2.7% from observed 0.685 |

Single residual correction: Barbero-Immirzi parameter  $\gamma \approx 0.2375$ , expected to be fixed by black hole entropy counting on the negotiation graph.

## 7 Document Series and Cross-References

This is Document 5 of 5 in The Medium series.

| Doc | File                         | Content                                                      |
|-----|------------------------------|--------------------------------------------------------------|
| 0   | medium_doc0_introduction.org | Personal introduction; the question behind the description   |
| 1   | medium_doc1_definition.org   | The Medium: Foundation Definition (primitives, three levels) |
| 2   | medium_doc2_unified.org      | Negotiation Modes, Particle Structure, Proton Mass           |
| 3   | medium_doc3_lagrangian.org   | Lagrangian of the Negotiation Field; QFT connection; SU(3)   |
| 4   | medium_doc4_hamiltonian.org  | Hamiltonian, energy spectrum, canonical quantization, ADM    |
| 5   | medium_doc5_G_H0.org         | <i>This paper</i>                                            |

All documents: Zenodo DOI: 10.5281/zenodo.18927154 · ORCID: 0009-0001-0328-2299 · [github.com/bkdepu/The-Medium](https://github.com/bkdepu/The-Medium)



## 8 References

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